

ISTANBUL TECHNICAL UNIVERSITY ★ CIVIL ENGINEERING FACULTY

**FLOOD HAZARD AND EXPOSURE ANALYSIS DESIGN WITH GIS FOR
KADIKOY DISTRICT**

DESIGN PROJECT THESIS

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JANUARY 2026

Beyza Bilge PAYAL, a bachelor student at Geomatics Engineering Department of İTÜ Civil Engineering Faculty with ID 010210631, successfully defended the thesis/dissertation entitled “ FLOOD HAZARD AND EXPOSURE ANALYSIS DESIGN WITH GIS FOR KADIKOY DISTRICT ”, which she prepared after fulfilling the requirements specified in the course description forms of Geomatics Engineering Design I and II courses, before the jury whose signatures are below.

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To our family,

FOREWORD

We take great pride and pleasure in presenting this design project, which marks the culmination of our undergraduate education at the Istanbul Technical University, Department of Geomatics Engineering. This study, titled "Flood Hazard and Exposure Analysis Design with GIS for Kadıköy District" has been an instructive academic journey for us.

We would like to express our deepest gratitude to our esteemed thesis advisor, Prof. Dr. Himmet KARAMAN, who has always stood by us with his knowledge and guidance throughout this process. The academic vision he provided not only shaped the methodology of this study but also inspired us to approach technical challenges with a scientific perspective.

We also have a particular obligation to express our gratitude to the friends we met during our Erasmus exchange program. The friendships and cultural interactions we shared broadened our horizons, contributing an international perspective to both our personal and academic growth.

Finally, we extend our endless thanks to our families for their unwavering patience, understanding, and moral support during this intense and demanding process.

January 2026

Beyza Bilge PAYAL & Ece KARAMAN

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ABBREVIATIONS

CN : Curve Number

DASK : Natural Disaster Insurance Institution (Doğal Afet Sigortaları Kurumu)

DEM : Digital Elevation Model

DSM : Digital Surface Model

GeoAI : Geospatial Artificial Intelligence

GIS : Geographic Information Systems

HEC-RAS : Hydrologic Engineering Center - River Analysis System

HWSD : Harmonized World Soil Database

IDF : Intensity-Duration-Frequency

ITU : Istanbul Technical University

LULC : Land Use / Land Cover

MCDA : Multi-Criteria Decision Analysis

OSM : OpenStreetMap

QGIS : Quantum Geographic Information System

RF : Random Forest

SCS : Soil Conservation Service

SCS-CN : Soil Conservation Service Curve Number

TWI : Topographic Wetness Index

UTM : Universal Transverse Mercator

WGS : World Geodetic System

SYMBOLS

A Cross-sectional area of flow (m^2)

CN Curve Number (dimensionless)

h Flood depth (m)

I_a Initial abstraction (mm)

n Manning's roughness coefficient

P Total precipitation / Rainfall (mm)

Q Direct surface runoff depth (mm)

Q_{peak} Peak discharge (m^3/s)

R Hydraulic radius (m)

S Potential maximum retention (mm)

S₀ Channel slope (m/m)

V Flow velocity (m/s)

α Specific catchment area (m^2/m)

β Local slope angle (rad)

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FLOOD HAZARD AND EXPOSURE ANALYSIS DESIGN WITH GIS FOR KADIKÖY DISTRICT

SUMMARY

Extreme precipitation driven by climate change and the increase in impermeable surfaces in rapidly urbanizing metropolises have significantly escalated urban flood risks. The district of Kadıköy, one of Istanbul's most densely populated areas, frequently experiences floods, particularly within the Kurbağalıdere basin and its vicinity. The primary objective of this senior design project is to conduct a high-resolution Flood Hazard and Exposure Analysis for the Kadıköy district using Geographic Information Systems (GIS) and analytical hydrological models, thereby eliminating dependency on commercial and complex hydraulic simulation software (e.g., HEC-RAS and MIKE FLOOD).

The study consists of three main stages. In the first stage, frequency analysis was performed on historical precipitation data using the Log Pearson Type III distribution. Peak discharge values were calculated for a 100-year period. Data was obtained from the DASK database (Station No: 35919). In the second stage, the Soil Conservation Service Curve Number (SCS-CN) method was employed to determine surface runoff potential. An average Curve Number (CN) for the basin was determined by considering Kadıköy's land use characteristics and soil properties. Additionally, the amount of effective rainfall transitioning into surface runoff was calculated.

Hydraulic modeling was carried out in the QGIS environment during the third stage. The Manning Equation, along with calculated peak discharges and roughness coefficients (n) specific to the urban texture, was utilized to determine water depths at critical points. Flood hazard maps, also known as flood inundation maps, were generated by processing water depth data onto a Digital Elevation Model (DEM). The exposure analysis aimed to identify structures and critical infrastructure elements under flood risk using OpenStreetMap (OSM) based building and road data.

Furthermore, to enhance the reliability of the study, the results of this physically-based model were compared with a "Flood Susceptibility Map" generated by the

Random Forest algorithm, a modern Artificial Intelligence (GeoAI) method. In conclusion, this project demonstrated that urban flood risks can be effectively modeled and suitable bases for decision-makers can be generated using engineering mathematics, open-source GIS tools, and data-driven approaches, without the need for expensive software.

KADIKÖY İLÇESİ İÇİN CBS İLE TAŞKIN TEHLİKE VE MARUZİYET ANALİZİ ÖZET

İklim değişikliğine bağlı aşırı yağışlar ve hızla kentleşen metropollerde geçirimsiz yüzeylerin artması, kentsel taşkın riskini önemli ölçüde artırmıştır. İstanbul'un en yoğun yerleşim bölgelerinden biri olan Kadıköy ilçesi, özellikle Kurbağalıdere havzası ve çevresinde sık sık taşkınlara maruz kalmaktadır. Bu bitirme tasarım projesinin temel amacı, Kadıköy ilçesi için yüksek çözünürlüklü bir Taşkın Tehlike ve Maruziyet Analizi yapmak için Coğrafi Bilgi Sistemleri (CBS) ve analitik hidrolojik modeller kullanarak ticari ve karmaşık hidrolik simülasyon yazılımlarına (örn. HEC-RAS ve MIKE FLOOD gibi) bağımlı olmamaktır.

Çalışma, üç ana aşamadan oluşmaktadır. Birinci aşamada, tarihsel yağış verileri üzerinde Log-Pearson Tip III dağılımı kullanılarak frekans analizi yapılmıştır. 100 ve tekerrür süresi için pik debi değerleri hesaplanmıştır. Veriler DASK veri tabanından (İstasyon No: 35919) elde edilmiştir. İkinci aşamada, yüzey akış potansiyelini belirlemek için Soil Conservation Service Curve Number (SCS-CN) yöntemi kullanıldı. Kadıköy'ün arazi kullanım özellikleri ve toprak özellikleri göz önünde bulundurularak havza genelindeki ortalama Eğri Numarası (CN) değeri bulunmuştur. Ayrıca, yüzeysel akışa geçen efektif yağışın miktarı hesaplanmıştır.

Hidrolik modelleme, üçüncü aşamada QGIS ortamında gerçekleştirilmiştir. Manning Denklemi ve hesaplanmış pik debiler, kentsel dokuya özgü pürüzlülük katsayıları (n) doğrultusunda kritik noktalardaki su derinliklerini belirlemek için kullanılmıştır. Taşkın tehlike haritaları olarak da bilinen taşkın yayılım haritaları, su derinliği verilerinin Sayısal Yükseklik Modeli (DEM) üzerine işlenmesiyle oluşturulmuştur. Maruziyet analizi, OpenStreetMap (OSM) tabanlı bina ve yol verilerini kullanarak taşkın riski altındaki yapılar ve önemli altyapı unsurları belirlemeyi amaçlamaktadır.

Ayrıca, çalışmanın güvenilirliğini artırmak için, fiziksel tabanlı bu modelin sonuçları, modern bir Yapay Zeka (GeoAI) yöntemi olan Rastgele Orman (Random Forest) algoritması tarafından oluşturulan "Taşkın Duyarlılık Haritası" ile karşılaştırılmıştır. Sonuç olarak, bu proje, pahalı yazılımlara ihtiyaç duymadan mühendislik matematiği, açık kaynaklı CBS araçları ve veri odaklı yaklaşımlar kullanılarak kentsel taşkın riskini etkin bir şekilde modellemenin ve karar vericiler için uygun altlıklar oluşturmanın mümkün olduğunu göstermiştir.

1. INTRODUCTION

Floods are a significant natural hazard that has a significant effect on urban environments and causes infrastructural, social, and economic disruptions. Density of urbanization and lower permeable surfaces have increased the area's vulnerability to floods in rapidly developing areas like Kadıköy. Traditional observation techniques often fail to capture flood susceptibility's complex spatial distribution in such dense urban fabrics. Consequently, the creation of accurate flood susceptibility maps has become an essential requirement for local decision-makers so that they can take preventative actions.

The Kadıköy region's flood susceptibility is the main objective of this study. The project's goal is to predict the spatial relationship between flood episodes and several environmental conditioning elements including elevation, slope, and land use by combining Geographic Information Systems (GIS) with sophisticated statistical analysis. To process these spatial variables and produce a precise susceptibility distribution map, the study used the Random Forest algorithm as a computational technique. The study's findings are meant to provide Istanbul with a trustworthy baseline map for upcoming infrastructure development and risk-reduction tactics.

1.1 Purpose of Thesis

This thesis's main goal is to create a thorough Flood Hazard Analysis for the Kadıköy area, focusing on the dangers brought on by the Kurbağalıdere basin's fast urbanization. The urgency of this study is underscored by real-world observations; as illustrated in Figure 1.1, recurring flash floods have frequently paralyzed the district's transportation network, necessitating a scientific risk assessment.



Figure 1.1 : Historical flash flood events in the study area (Kadıköy).

By combining analytical hydrological models with Open Source Geographic Information Systems (QGIS), this study seeks to demonstrate an economical and scientifically sound methodology, in contrast to conventional approaches that mostly rely on sophisticated commercial hydraulic software like HEC-RAS. To determine the peak discharge value over a 100-year return period based on historical precipitation data, the study will first conduct a hydrological frequency analysis using the Log-Pearson Type III distribution.

The surface flow potential is then estimated using the SCS-CN (Soil Conservation Service Curve Number) approach, and flood depths are analytically modeled without hydrodynamic simulations using Manning's Equation. Cross-validating these physical-based results with a data-driven Random Forest (GeoAI) susceptibility model is a crucial goal of this study in order to evaluate the dependability of the suggested methodology. Lastly, the study hopes to provide useful risk assessments for urban planning and decision-making by identifying the critical flood-prone zones.

1.2 Scope of Thesis

The administrative limits of Istanbul's Kadıköy District serve as the study's geographical boundaries. To reduce distortion in area and distance calculations, all

spatial analyses are standardized to the EPSG:32635 (WGS 84/UTM Zone 35N) projection system. Using a 100 year return period design storm based on the maximum 24-hour rainfall data ($P = 103.91\text{mm}$) acquired from the MGM Göztepe Station, the hydrological scope is rigorously restricted to the examination of pluvial floods.

In terms of methodology, the study carefully avoids using commercial hydrodynamic simulation software like HEC-RAS in favor of a GIS-based approach combined with the SCS-CN (Soil Conservation Service-Curve Number) method to predict surface runoff. Rather, the study uses a customize Peak Discharge Depth Calculator macro for the analytical computation of flood depths based on Manning's Equation and QGIS 3.44.6 and GRASS GIS algorithms (particularly r.watershed for catchment delineation with a 500-pixel threshold) for spatial processing. The final analysis's scope includes creating classified flood hazard maps and evaluating exposure by superimposing these hazard layers on top of the DASK building inventory to pinpoint the district's most susceptible buildings and residents.

1.3 Literature Review

The literature often divides flood hazard mapping and susceptibility assessment research into two categories: GIS based conceptual models and physically based hydrodynamic models. This section evaluates the mathematical underpinnings and capabilities of popular flood analysis tools and approaches.

1.3.1 Hydrodynamic and Hydrologic Modeling Software

1.3.1.1 HEC-RAS (Hydrologic Engineering Center's River Analysis System)

HEC-RAS, created by the US Army Corps of Engineers (USACE), is regarded as the industry standard for simulating river hydraulics. One dimensional (1D) and two-dimensional (2D) flow simulations are supported by the program. The Continuity Equation (Mass Conservation) and the Momentum Equation serve as the foundation for the software's computational engine. It uses finite difference techniques to solve the St. Venant Equations for unstable flow situations and the Energy Equation (based on the Bernoulli principle) for steady flow assessments. Although recent studies like

Peker et al. (2024) show that HEC-RAS is successful in producing detailed hydraulic outputs (water depth, velocity vectors), its application in data-scarce places is difficult due to its complex interface and requirement for detailed cross-section data.

1.3.1.2 MIKE FLOOD

The comprehensive modeling program MIKE FLOOD, created by DHI, dynamically links 1 dimensional river models (MIKE 11) with 2 dimensional floodplain models (MIKE21).

The model simulates intricate flows in both urban and rural settings by solving the depth-averaged Shallow Water Equations (Navier-Stokes). The capacity of MIKE FLOOD to mimic the water transfer between the riverbed (1D) and the floodplain (2D) utilizing techniques like "Lateral Link" and "Standard Link" based on physical principles is its most notable feature. However, its broad application in academic studies is significantly limited by hefty licensing prices and computational constraints.

1.3.1.3 SWAT (Soil and Water Assessment Tool)

SWAT is a physically based model that simulates hydrological processes at the catchment scale over extended periods of time. It is especially used to examine how changes in land use affect the water balance. SWAT calculates surface runoff using the SCS-CN (Soil Conservation Service Curve Number) method, which is also the basic approach used in this thesis. By splitting the watershed into Hydrological Response Units (HRUs), it effectively establishes the infiltration and runoff balance.

1.3.2 Data driven and GIS Based Approaches

Because complicated hydrodynamic models demand large amounts of data and processing power, researchers have turned to more useful techniques that are integrated with Geographic Information Systems (GIS). By combining spatial data (DEM) with empirical formulas like Manning's Equation, open-source programs like QGIS and GRASS GIS may effectively determine "steady-state" flood depths. This approach provides an affordable substitute, particularly for

urban planning studies that require high geographic precision but not the high temporal resolution of software such as HEC-RAS.

In recent years, the use of Machine Learning algorithms (e.g., Random Forest) has increased to test the accuracy of physical models. These methods play a complementary role in generating flood susceptibility maps by learning the non-linear relationships between parameters such as slope, elevation, and TWI (Topographic Wetness Index).

2. STUDY AREA AND DATASET

2.1 Geographical Setting

The Kadıköy District, situated on Istanbul's Anatolian side at the southernmost point of the Istanbul Strait, is the subject of the study. The district is roughly 25.2km² in size and is situated between longitudes 29°01'–29°04' East and latitudes 40°58'–41°00' North. It shares borders with the Sea of Marmara to the south and west, Ataşehir to the northeast, Maltepe to the east, and Üsküdar to the north. (Figure 2.1)

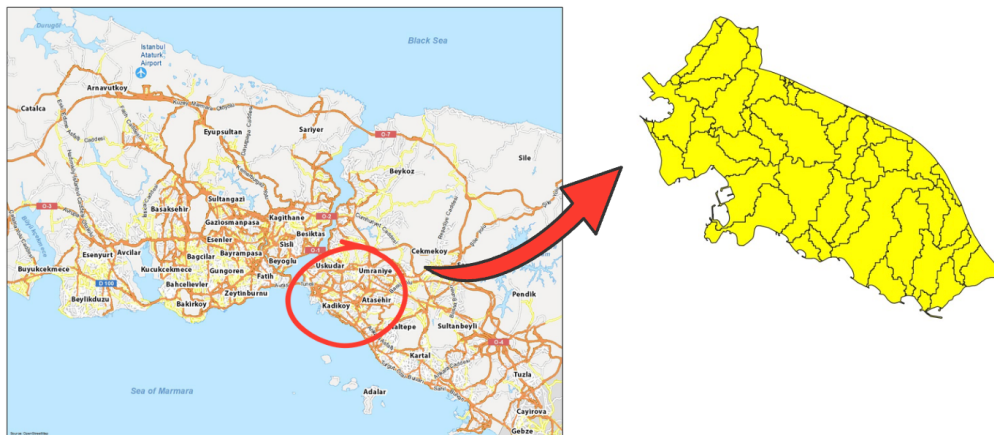


Figure 2.1 : Study Area

2.2 Topographic and Hydrological Characteristics

Low-lying slopes that descend into the Sea of Marmara are a common feature of Kadıköy's terrain. The Kurbağalıdere Stream, the district's most important hydrological feature, serves as the primary drainage basin, collecting surface runoff from the nearby districts (Ataşehir and Ümraniye) and releasing it into the Marmara Sea (Moda Bay). Due to the Kurbağalıdere valley's relatively low elevation and the "bowl-shaped" terrain of the center neighborhoods, the area is inherently vulnerable to water buildup during periods of intense rainfall.

2.3 Urbanization and Land Use

Kadıköy, being one of the most established districts of Istanbul, exhibits a highly dense urban texture . The area is characterized by widespread impervious surfaces, including extensive roof coverage, concrete, and asphalt infrastructure. This intense urbanization severely reduces the soil's natural infiltration capacity, leading to rapid surface runoff generation during heavy rainfall events. Consequently, the district is highly susceptible to "Urban Flash Floods," underscoring the necessity of a comprehensive flood hazard assessment and model validation study.

Land use information was obtained from the CORINE Land Cover database in order to calculate the surface runoff potential using the SCS-CN method. The spatial information required to classify Kadıköy's urban fabric is provided by this dataset.

2.3.1 Reclassification of Land Use and Land Cover (LULC) Data

The raw Land Use and Land Cover (LULC) datasets obtained for Kadıköy were reclassified to better represent the district's unique urban morphology and to provide a standardized input for the hydrological modeling process. This reclassification groups different urban fabrics based on their physical characteristics, building density, and surface imperviousness. Table 2.1 summarizes these categories along with local examples from Kadıköy, serving as the basis for assigning Curve Numbers (CN) and Manning's roughness coefficients in the subsequent analysis stages.

Table 2.1 : Land Use/Cover Classification Categories Adopted for the Study Area

Land Use Category	Description & Characteristics	Typical Examples in Kadıköy
High Density Residential	Continuous urban fabric; attached buildings with very limited open space. High imperviousness (>90).	Osmanağa, Caferağa (Moda), Rasimpaşa (Yeldeğirmeni)
Medium Density Residential	Discontinuous urban fabric; apartment blocks with gardens or green gaps between structures.	Erenköy, Suadiye, Fenerbahçe
Commercial & Public Units	Large-scale structures including shopping malls, hospitals, universities, and business centers.	Tepe Nautilus, Marmara Univ. Campus, Palladium Area
Transportation Networks	Major highways, arterial roads, intersections, and open parking lots. 100% impervious.	D-100 (E-5) Highway, Baghdad Avenue, Minibus Road
Urban Green Areas	Recreational parks, groves, cemeteries, and sports facilities. Permeable surfaces.	Validebağ Grove, Yoğurtçu Park, Karacaahmet Cemetery
Water Bodies	Open water surfaces (streams, pools, sea).	Kurbağalıdere Stream Channel, Kalamış Marina

2.4 Dataset

In order to conduct hydrological modeling and flood exposure assessments for the Kadıköy district, a multi-source geospatial database was built. The collection includes infrastructure inventory, topography information, land use characteristics, and weather records. All spatial datasets were re-projected to the WGS 84 / UTM Zone 35N (EPSG:32635) coordinate system in order to guarantee spatial consistency throughout computations (area, distance, and overlay analysis).

2.4.1 Meteorological Data

The Turkish State Meteorological Service (MGM) provided historical precipitation data. In particular, the study area was spatially represented by data from the Göztepe Meteorology Station (Station No: 35919). The yearly maximum daily rainfall records were subjected to a frequency analysis in order to calculate the design storm strength. Based on the Log Pearson Type III distribution, the maximum 24 hour rainfall depth for the 100 year return period (Q_{100}) was determined to be $P = 103.91\text{mm}$.

2.4.2 Topographic and Soil Data

Digital Elevation Model (DEM): A 30-meter resolution DEM from the Copernicus Global DSM was used as the primary input for topographic analysis. This raster data was pre-processed in GRASS GIS (using the *r.watershed* algorithm) to delineate micro-catchments and extract drainage networks.

Soil Characteristics: Soil permeability data was obtained to identify the Hydrologic Soil Groups (HSG). Based on the geological structure of Kadıköy, Group D soils (characterized by low infiltration and high runoff potential) were identified as the dominant soil type.

2.4.3 Exposure Data

For the exposure and susceptibility assessment phase, building footprints and road network layers were compiled from OpenStreetMap (OSM) vectors.

The inventory of the datasets used in this study is summarized in Table 2.1.

Table 2.2 : Datasets

Data Category	Data Type	Source / Resolution	Application Purpose
Topography	Raster (DEM)	Copernicus / 30m	Catchment delineation, Slope, and TWI calculation.
Meteorology	Time Series	MGM (Göztepe St.)	Calculation of 100-year rainfall depth (P=103.91 mm).
Land Cover	Vector/Raster	CORINE / 1:100.000	Determination of Manning's n and Curve Numbers (CN).
Hydrology	Raster Grid	Derived (SCS-CN)	Estimation of surface runoff (Q) and infiltration (S).

3. METHODOLOGY

The comprehensive methodological framework of the study, from data collection to model validation, is described in the process shown in Figure 3.1. The Copernicus DEM and CORINE datasets are processed first, and then physical flood depths (h) are calculated using Manning's Equation and the SCS-CN method. Lastly, the Random Forest (GeoAI) method is used to cross-validate the produced Flood Hazard Maps in order to guarantee spatial consistency and dependability.

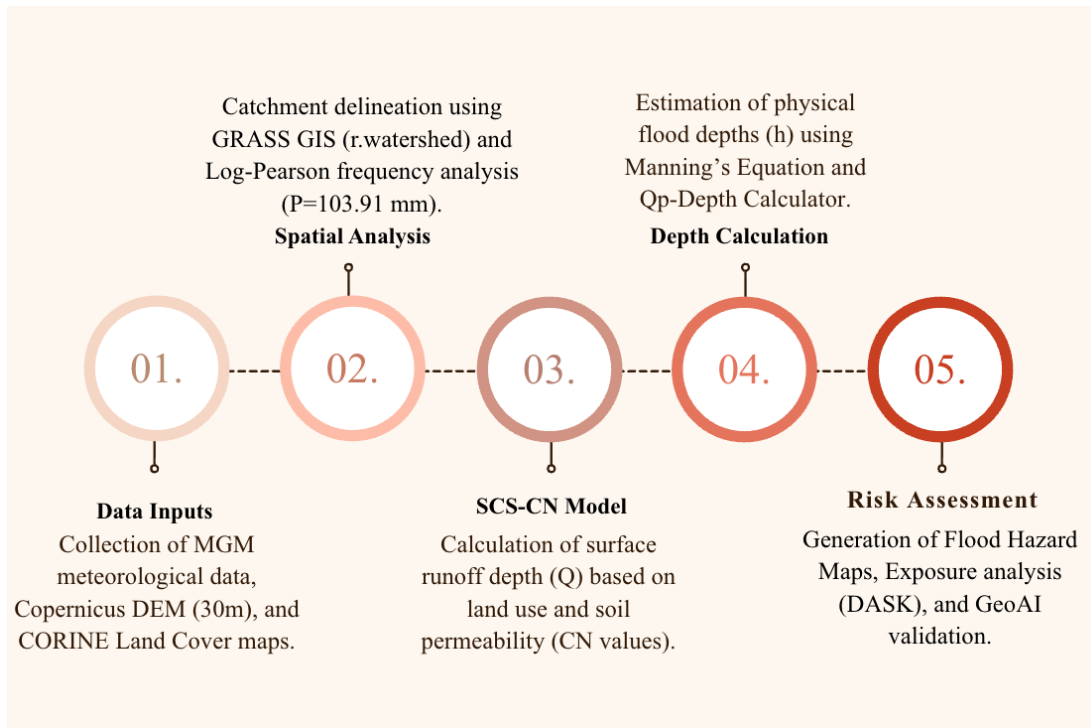


Figure 3.1 : Flowchart of the study.

This study's process is organized around a GIS-based analytical framework that combines hydraulic depth estimation, surface runoff modeling, and hydrological frequency analysis. Hydrological analysis, catchment delineation, surface runoff calculation (SCS-CN), and hydraulic depth estimation (Manning) are the four successive steps of the process.

3.1 Hydrological Frequency Analysis

A statistical frequency study of historical precipitation data from the Göztepe Station (No: 35919) was used to calculate the design storm strength. The peak discharge for a 100 year

return period (Q_{100}) was computed using the Log-Pearson Type III distribution, the conventional statistical method for flood frequency prediction. In this case, the computed maximum 24 hour rainfall depth (P) is 103.91 mm.

3.2 Catchment Delineation with GRASS GIS

For urban flood modeling, a precise characterization of drainage basins is essential. The 30m DEM was processed using the GRASS GIS program, specifically the r.watershed method. The Kadıköy district's micro catchments were identified using a drainage threshold of 500 pixels. The main spatial units for the runoff computation were then created by vectorizing and clipping these raster catchments to the district boundaries.

3.3 Surface Runoff Modeling (SCS-CN Method)

The direct surface runoff (Q) from the rainfall (P) was estimated using the USDA-developed Soil Conservation Service Curve Number (SCS-CN) approach. This approach uses the Curve Number (CN) parameter to account for soil permeability and land use variables. Initially, each pixel's Potential Maximum Retention (S) was determined using the following formula:

$$S = \frac{25400}{CN} - 254$$

The 100-year rainfall depth (P = 103.91 mm) was then used to calculate the direct surface runoff depth (Q):

$$Q = \frac{(P - 0.2S)^2}{P + 0.8S}$$

(Conditions where $P > 0.2S$ were considered for runoff generation).

Subsequently, the calculated runoff depth (Q_{depth} in mm) was converted into volumetric peak discharge (Q_{peak} in m^3/s) for each micro-catchment. This conversion was achieved by multiplying the runoff depth by the catchment area ($A_{\text{catchment}}$) and applying a unit conversion factor, assuming the runoff occurs effectively within the concentration time of the basin.

3.4 Hydraulic Depth Estimation (Manning's Equation)

In contrast to unsteady flow simulations (like HEC-RAS), this work estimates maximum flood depths using a steady-state analytical method. Manning's Equation was solved iteratively for each catchment using a specially designed "Qp-Depth Calculator" macro. For the hydraulic geometry, each micro-catchment's outlet was conceptualized as a rectangular open channel with a representative width (w) of 5.0 meters. This assumption allows for the calculation of the Cross-sectional Area ($A = w \times y$) and Wetted Perimeter ($P = w + 2y$) required for the iterative solution. The equation used is:

$$Q = \frac{1}{n} \cdot A \cdot R^{\frac{2}{3}} \cdot S_0^{\frac{1}{2}}$$

Q: Peak discharge (m³/s) derived from the SCS-CN method.

n: Manning's roughness coefficient (derived from LULC map).

A: Cross-sectional area of the flow.

R: Hydraulic radius.

S₀ : Slope of the terrain.

The calculated depth values (h) were then joined to the spatial vector layers in QGIS to generate the final Flood Hazard Map.

The hydraulic behavior of the surface runoff was modeled by assigning specific Manning's roughness coefficients (n) to each land use class, as detailed in Table 3.1. These values represent the flow resistance encountered as water travels across different surface materials in Kadıköy, directly influencing the calculated flow velocities and inundation depths.

Table 3.1 : Manning’s Roughness Coefficients (n) Adopted for Hydraulic Modeling

Surface Type	Manning’s (n) Value	Description / Justification
Impervious Surfaces	0.013 - 0.015	Smooth asphalt roads and concrete channels; allows maximum flow velocity.
Urban Ground	25	Compacted soil, pavements, and residential gaps within dense urban fabric.
Natural Channels	35	Irregular natural stream beds (e.g., Kurbağalıdere sections) with vegetation and debris.
Dense Vegetation	50	Parks and groves (e.g., Yoğurtçu Park or local gardens); high resistance to flow.

4. RESULTS AND ANALYSIS

4.1 Hydrological Model Results and Surface Runoff Analysis

The surface runoff potential of the Kadıköy district was quantitatively evaluated using the SCS-CN method. Significant spatial variability was observed based on the Land Use/Land Cover (LULC) data.

High Runoff Zones: In the commercial and dense residential areas, particularly in the eastern parts of Osmanağa (Kurbağalidere basin) and Caferağa, Curve Number (CN) values reached 95. Due to the high percentage of impervious surfaces (concrete/asphalt) in these locations, approximately 90% of the rainfall is converted into surface runoff.

Low Runoff Zones: Conversely, recreational areas such as the coastal green belt and Yoğurtçu Park exhibited CN values between 60 and 70, indicating a significantly higher retention capacity.

Model Results: Under the calculated 100 year rainfall intensity ($P=103.91$), the spatial analysis revealed that the peak accumulated runoff volumes reached 2.13 million m^3 in the impervious urban core. In contrast, green zones showed significantly lower accumulation values, ranging between 0 – 0.43 million m^3 .

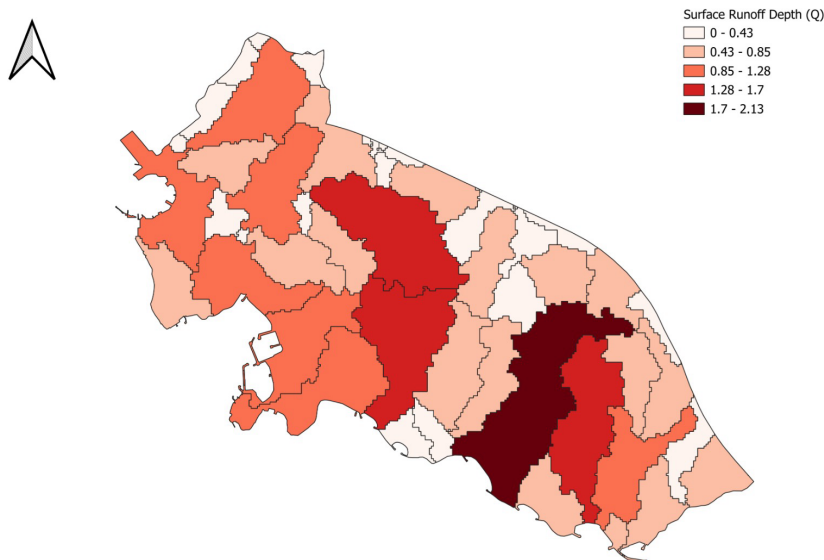


Figure 4.1 : Surface Runoff (Q) Map

4.2 Land Use and Distribution of CN Values

Significant variations in the hydrological response of the study area were found when the relationship between Land Use/Land Cover (LULC) and soil permeability was investigated.

Curve Number (CN) values reached levels of 95 in commercial areas and dense residential zones, especially in the eastern parts of Osmanağa and Caferağa neighborhoods where the rate of urbanization is considerable. This shows that without penetrating the soil, almost 90% of the rainfall in these areas is immediately transformed into surface runoff.

On the other hand, recreational places along the coastal belt and green spaces like Yoğurtçu Park and Özgürlük Park were found to have CN values between 60 and 70. These regions exhibit low runoff coefficients and a high retention capacity.

According to the LULC research, Kadıköy's predominant land use class is Continuous Urban Fabric (111). Impervious surfaces significantly grow as a result of this high degree of urbanization. As a result, the district is considered a high-risk zone for urban floods due to the high Curve Number (CN) values found in these regions, which suggest a significant potential for surface runoff.

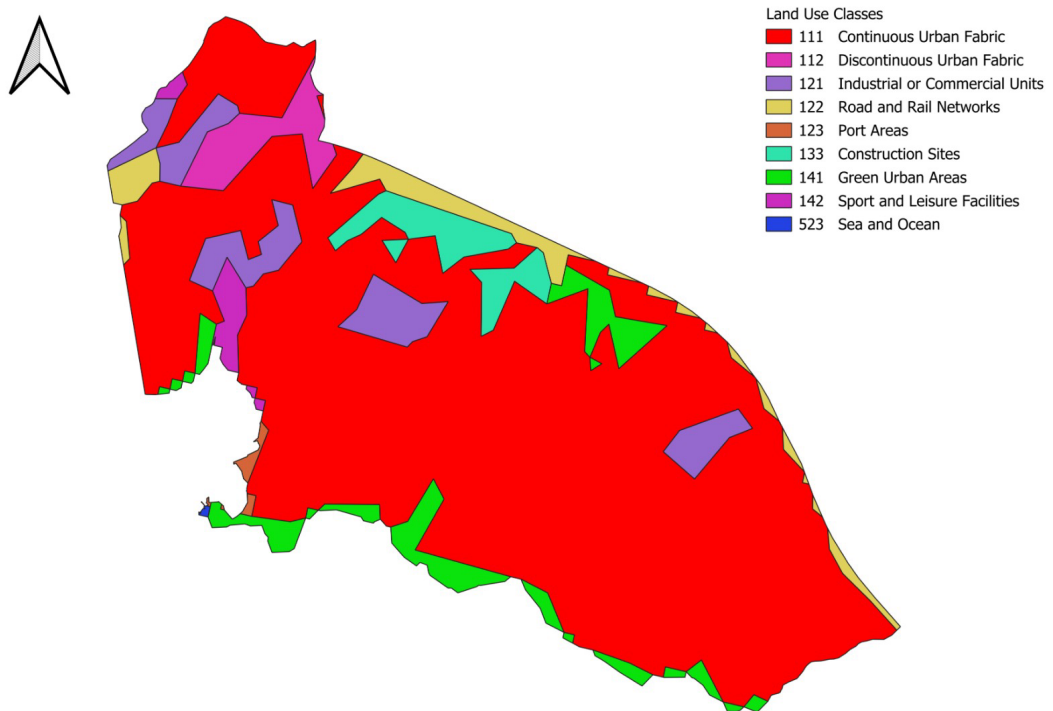


Figure 4.2 : Land use classification map

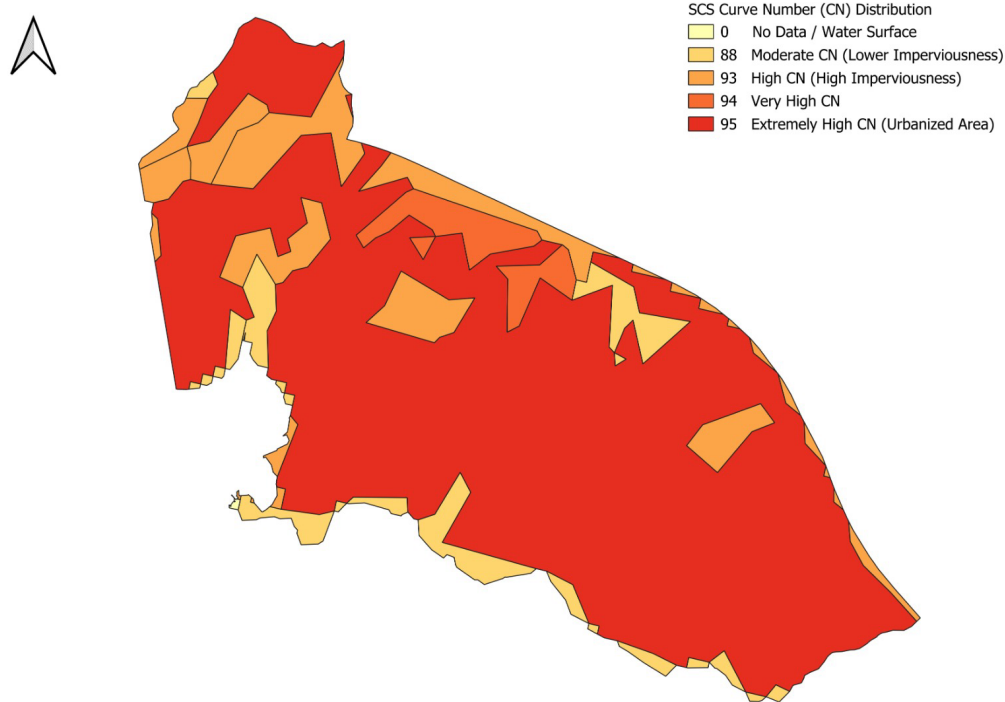


Figure 4.3 : CN Map

As detailed in Table 4.3 , high Curve Number (CN) values (>90) correspond to densely urbanized zones and transportation networks where soil permeability is nearly zero. Conversely, lower CN values (<79) indicate green areas such as Validebağ Grove, which act as natural sponges, facilitating water infiltration and reducing surface runoff.

Table 4.1 : CN Values Used in the Kadıköy Susceptibility Map and Corresponding Land Cover Classes

Map Legend Value (CN)	Land Cover Definition	Physical Description	Permeability & Runoff Status
98 - 100	Transportation & Hard Surfaces	Asphalt roads, parking lots, E-5 Highway.	Fully Impervious (100% Runoff)
90 - 95	Dense Urban Fabric	Attached buildings (e.g., Osmanağa, Caferağa), commercial areas.	Very High Runoff Potential
80 - 89	Discontinuous Urban Fabric	Gated communities, detached houses with gardens, schoolyards.	Moderate Runoff Permeable
60 - 79	Green Spaces	Urban parks, groves (e.g., Validebağ), cemeteries.	(High Infiltration) No Infiltration / Storage
< 60	Water Bodies	Open water surfaces, pools.	Capacity

The CN map (Figure 4.3) shows that Kadıköy has a high potential for runoff, with most values falling between 93 and 95. These figures show a very impermeable watershed with a very low potential maximum retention (S) of the soil. The district's hydrologic risk is mostly driven by urban density, as confirmed by the spatial distribution.

In Kadıköy, a substantial geographical association between urban density and hydrological risk is confirmed by the combined study of LULC and SCS Curve Number (CN), where regions designated as Continuous Urban Fabric (111) directly correspond to extreme CN values of 95. This relationship shows how the district's low-permeability Group D soils and high degree of artificial surface sealing have significantly decreased the natural infiltration capacity (S), enhancing the potential for surface runoff. Because of a shorter duration of concentration, the occurrence of very high CN values (93–95) suggests that the catchment will react quickly to heavy precipitation, greatly increasing the vulnerability to flash floods.

4.3 Spatial Distribution of Risk and 3D Evaluation

A dual visualization technique using the 2D Spatial Distribution Map (Figure 4.4) and the 3D Volumetric Model (Figure 4.5) is used to thoroughly assess the Flood Risk Score, which is the accumulated water depth obtained from the hydraulic model. The 2D map shows that the horizontal extent of flood susceptibility is not evenly distributed but is concentrated in certain hydromorphological zones, such as the Kurbağalıdere Basin and the eastern portions of Osmanağa, where critical risk corridors created by the combination of low elevation and high imperviousness are clearly delineated by dark blue gradients.

In addition to this spatial analysis, the 3D model extrudes these susceptibility scores vertically to show the amount of water buildup, demonstrating that the flood susceptibility is a serious volumetric hazard rather than just a surface problem. In contrast to the low danger seen on the high-elevation ridges of Acıbadem, the elevated blue blocks represent topographic "sink points" where runoff physically accumulates and may surpass surface drainage capacity, especially in the valley bottoms. As a result, our comprehensive study shows that the "bowl effect" of the topography in conjunction with dense urbanization is the primary cause of the flood susceptibility,

indicating these volumetric hotspots as the most vulnerable locations in need of focused drainage solutions.

The flood susceptibility classification presented in Table 4.1 utilizes a monochromatic blue color scale to visually represent the increasing intensity of water depth. The intervals are categorized into 5 distinct classes based on the Natural Breaks (Jenks) method. Darker shades of blue correspond to higher water depths, with the deepest shade (Navy/Deep Blue) highlighting the most critical zones where the water depth exceeds 0.18 m, such as the Osmanağa depression.

Table 4.2 : Flood Susceptibility Assessment Matrix and Classification Scale (Blue Gradient)

Susceptibility Level	Risk Score / Water Depth (h) Range	Color Code (Map)	Potential Impact Description
Very Low Susceptibility	0.00 - 0.04	Very Light Blue (Transparent)	Negligible surface runoff; safe for daily life.
Low Susceptibility	0.04 - 0.09	Light Blue	Minor water accumulation; generally safe.
Moderate Susceptibility	0.09 - 0.13	Medium Blue	Water depth reaches ankle level; pedestrian movement slows down.
High Susceptibility	0.13 - 0.18	Dark Blue	Water reaches curb height; traffic disruption likely.
Very High Susceptibility	0.18 - 0.23	Navy / Deep Blue	Critical Threshold. Susceptibility of structural damage and basement flooding (Max observed: 0.23).

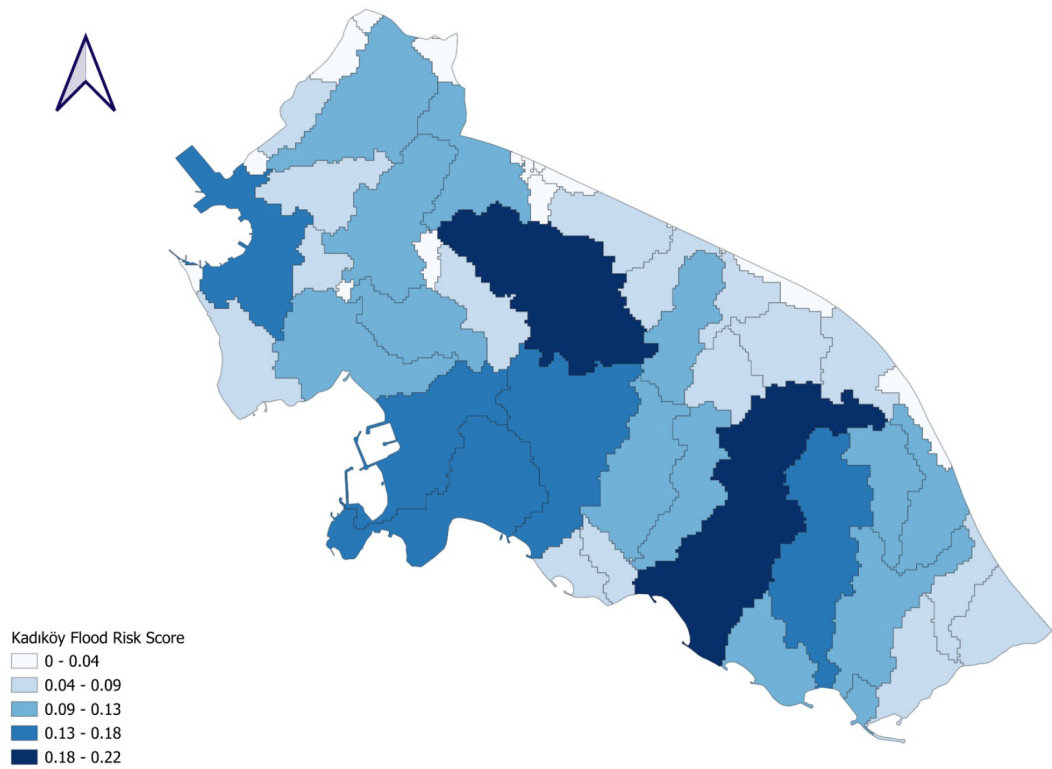


Figure 4.4 : Flood Susceptibility Score Map in 2D

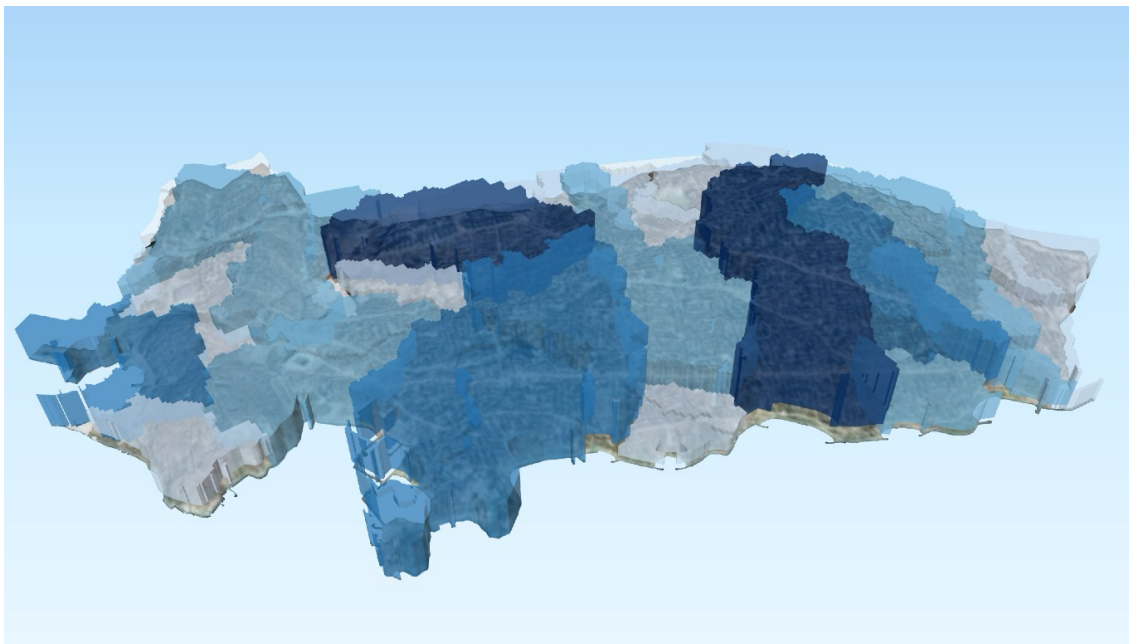


Figure 4.5 : Flood Susceptibility Score Map in 3D

Upon examining the Table 4.1 , it is observed that while the water depth reaches 0.23 meters in sub-basin 614 (Very High Susceptibility), it remains minimal between 0.01 and 0.03 meters in sub-basins 472 and 476 (Low susceptibility). Specifically, the high slope value of 7.33% in sub-basin 476 confirms that this area functions as a runoff generator, allowing water to drain rapidly without accumulation, thereby preventing flood susceptibility.

Table 4.3 : Computed Peak Discharge (Q_{peak}) and Water Depth (y) Results for Selected Sub-basins

Sub-basin ID	Slope (%)	Peak Discharge (Q_{peak}) (m^3/s)	Calculated Water Depth (y) (m)	Susceptibility Assessment
614	3.55 %	2.13	0.23	Very High Susceptibility
594	4.05 %	1.67	0.19	High Susceptibility
604	3.49 %	1.46	0.18	High Susceptibility
646	3.64 %	1.34	0.17	High Susceptibility
602	3.70 %	1.10	0.15	Medium-High Susceptibility
600	3.66 %	0.99	0.14	Medium Susceptibility
432	3.77 %	0.97	0.14	Medium Susceptibility
460	4.22 %	0.91	0.13	Medium Susceptibility
476	7.33 %	0.08	0.03	Low Susceptibility (Safe)
472	4.13 %	0.02	0.01	Low Susceptibility (Safe)

The spatial distribution of flood susceptibility, when analyzed alongside neighborhood boundaries, reveals significant vulnerabilities in the district's urban core. Specifically, the Osmanağa and Caferağa neighborhoods, characterized by high-density urban fabric and asphalt infrastructure, correspond to the highest Curve Numbers (CN: 90–95), where nearly 90% of rainfall converts directly into surface runoff. The 3D volumetric evaluation further demonstrates that these neighborhoods act as "topographic traps" within the Kurbağalıdere Basin, where runoff from higher elevations like Acıbadem physically accumulates due to the bowl shaped terrain and low lying slopes. In contrast, neighborhoods containing significant green spaces, such as those encompassing Validebağ Grove or Yoğurtçu Park, exhibit lower susceptibility scores (CN: 60–79), confirming their role as natural sponges that alleviate the hydraulic load on the urban drainage network.

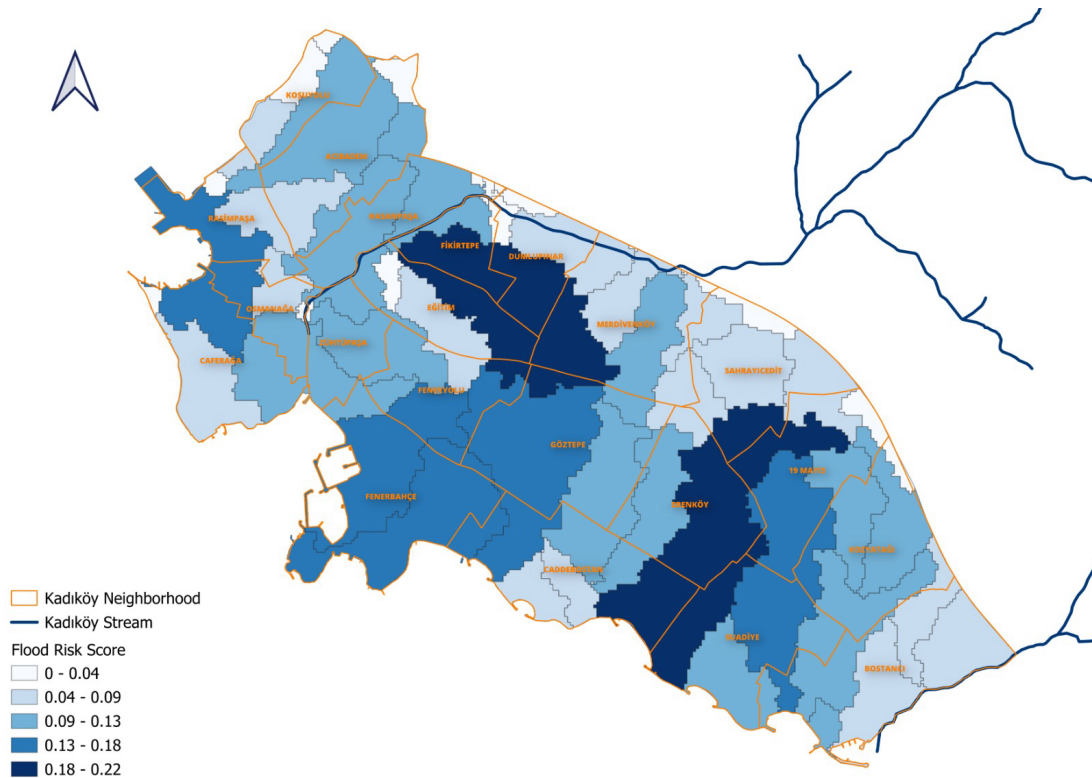


Figure 4.6 : Figure Flood Susceptibility Map in 2D

4.4 Comparative Study: GeoAI Driven Susceptibility Mapping

The flood susceptibility of the Kadıköy district was cross-validated using a data driven method in addition to physical-based hydrological modeling (SCS-CN and Manning's Equation). A GeoAI based flood susceptibility mapping investigation using the Random Forest (RF) machine learning technique is presented in this section. This comparison analysis aims to assess the degree of consistency between data-driven artificial intelligence models and physically based hydraulic models.

4.4.1 Methodology of the GeoAI Model

The GeoAI framework was built upon open-source geospatial data. A 30 m resolution Digital Elevation Model (DEM) from the Copernicus Global dataset was utilized as the primary input. The workflow consisted of the following steps:

4.4.1.1 Feature Engineering:

Topographic conditioning factors, specifically Elevation, Slope, and Topographic Wetness Index (TWI), were extracted using QGIS and SAGA GIS tools. TWI was calculated to identify water accumulation zones using the formula:

$$TWI = \ln\left(\frac{\alpha}{\tan \beta}\right)$$

where α is the specific catchment area and β is the local slope angle.

4.4.1.2 Inventory Preparation:

A balanced dataset comprising 30 ground truth inventory points (representing both flood-prone and Susceptibility free locations) was created to train the model.

4.4.1.3 Model Training:

A Random Forest classifier was trained to learn the non-linear relationships between topographic features and flood occurrences.

4.4.2 Results of the GeoAI Assessment and Comparative Validation

For the hydrological outcomes of the current investigation, the GeoAI model's predicted performance served as an essential cross validation tool. Since the two methodologies follow fundamentally different technical approaches where the SCS-CN and Manning-based model relies on deterministic hydraulic physics and the Random Forest model operates on stochastic data driven patterns a visual comparative analysis was performed to evaluate their spatial consistency. The most important discovery is the strong spatial convergence between the two approaches: as shown in Figure 4.6, the "High Susceptibility" clusters (Index > 0.8) predicted by the AI model and the "High Hazard" zones determined using Manning's Equation in this project significantly overlap. The primary objective of this visual comparison is not a direct numerical correlation, but rather the validation of identified high risk clusters across different modeling paradigms. This consistency demonstrates that the flood hazard map generated in this study is physically based and trustworthy, as its conclusions are supported by an independent, data driven artificial intelligence algorithm. This multi model visual cross validation provides a higher degree of reliability for decision makers by ensuring that critical hotspots, such as the Osmanağa and Kurbağalıdere corridors, are consistently identified regardless of the underlying algorithm.

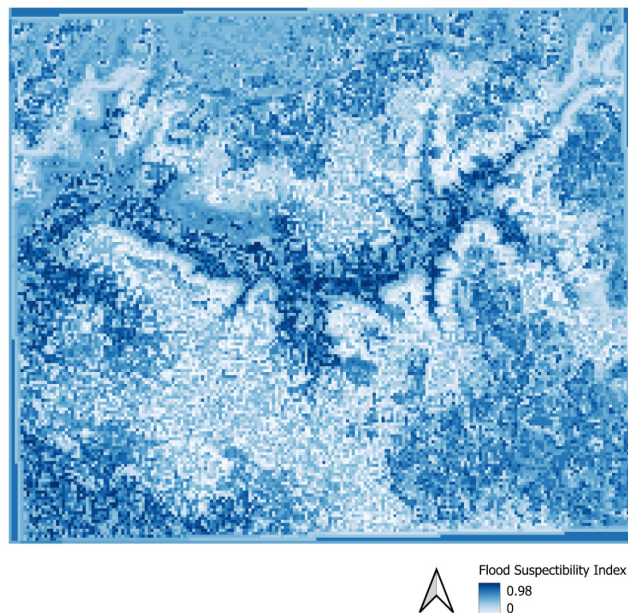


Figure 4.7 : The Flood Susceptibility Map generated by the Random Forest (GeoAI) algorithm for cross validation.

The GeoAI based deep learning model and the conventional physical hydrological approach have a strong spatial association, according to a comparison of flood susceptibility in the Dumlupınar, Merdivenköy, and Sahrayıcedit neighborhoods. The GeoAI technique offers a high-resolution, pixel-based probability gradient that captures intricate non-linear spatial patterns, whereas the physical model classifies risk based on deterministic factors like slope, Curve Number (CN), and Land Use/Land Cover (LULC). The peak probability areas in the GeoAI output (scores up to 0.98) and the high susceptibility zones identified by the physical model (scores ranging from 0.18 to 0.22) are significantly aligned in all three pilot areas, especially along the topographic depressions of Dumlupınar and the southern corridors of Sahrayıcedit. This alignment provides a reliable and computationally efficient decision support tool for urban flood management in Kadıköy, confirming that machine learning models can accurately replicate actual flood realities.

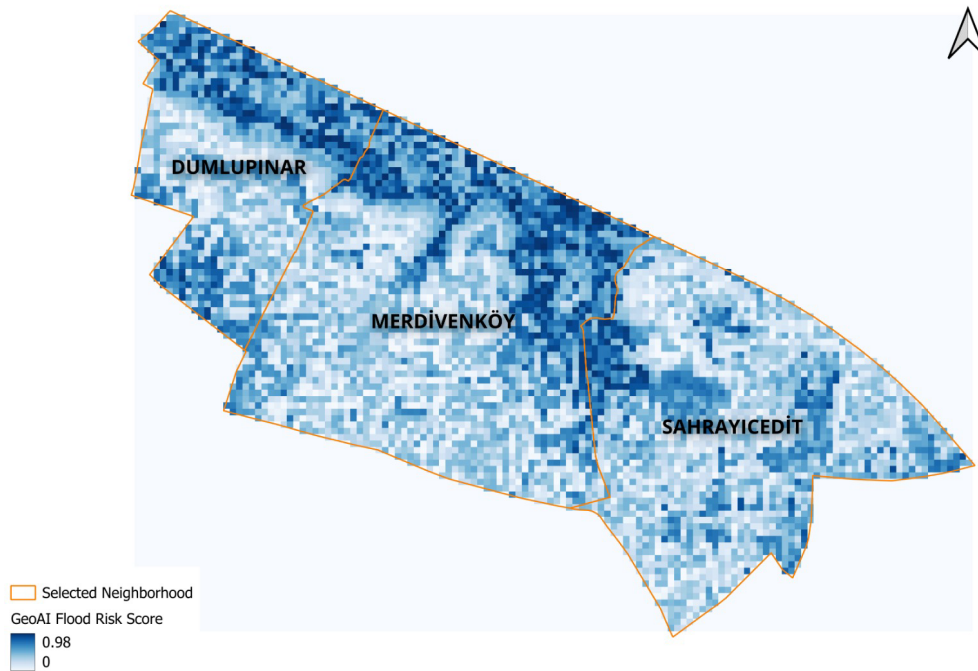


Figure 4.8 : Selected Neighborhoods in The Flood Susceptibility Map generated by the Random Forest (GeoAI).

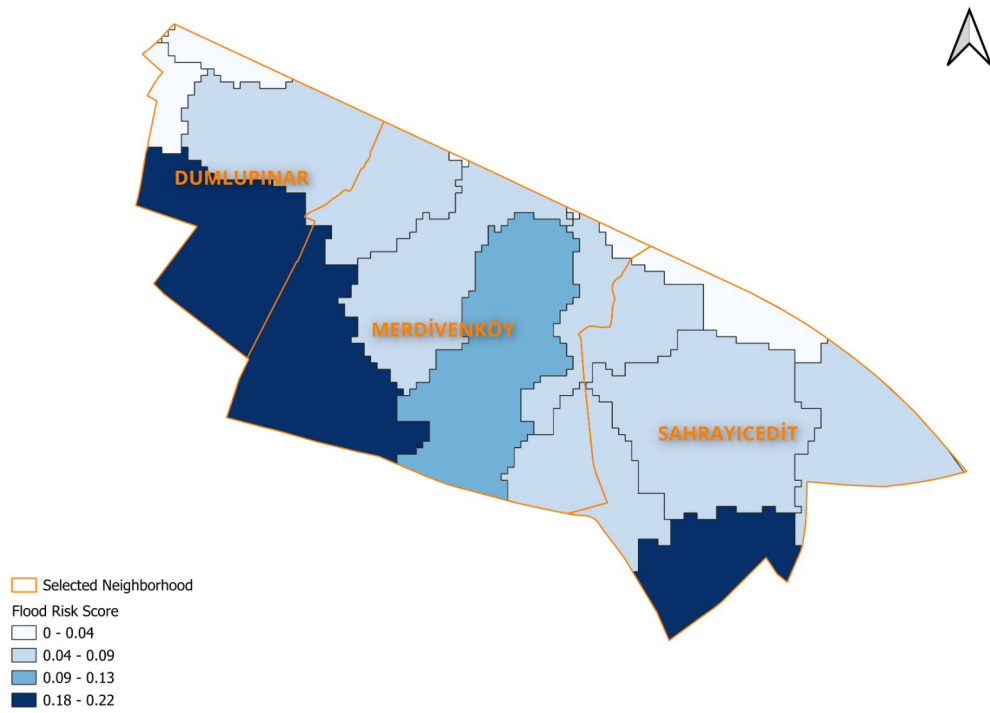


Figure 4.9 : Selected Neighborhoods in The Flood Susceptibility Map

5. CONCLUSIONS

In this study, a comprehensive flood susceptibility analysis was conducted for the Kadıköy basin, one of the most densely urbanized and strategically important districts of Istanbul, by integrating Geographic Information Systems (GIS) and hydraulic modeling techniques. The methodological originality of the study lies in the processing of static terrain data (CN, Slope, Area) and dynamic rainfall data through macro algorithms, thereby modeling not only the location of the risk but also its physical magnitude.

In the analysis process, the surface runoff potential of the basin was calculated using the SCS-CN (Soil Conservation Service Curve Number) method, while the physical water depths (y) and flow volumes in the field were numerically revealed using Manning's Hydraulic Equation. The analysis results confirm the critical and direct impact of Land Use/Land Cover (LULC) decisions on flood formation. Specifically, in the Osmanağa and Caferağa neighborhoods, characterized as the urban core with dense adjacent buildings and asphalt surfaces, CN values were found to reach the 90-95 band. It was observed that in these regions dominated by impervious surfaces, the water retention capacity of the soil drops to almost zero, and more than 90% of the precipitation falling in a 100 year recurrence interval scenario converts directly into surface runoff without infiltrating the soil. This situation creates a sudden and intense hydraulic load on the urban infrastructure.

As a result of the simulations, it was calculated that peak discharge values (Q_{100}) reached levels $2.129 \text{ m}^3/\text{s}$ in the aforementioned impervious zones, and with the hydraulic load created by this discharge, water depths could rise up to 0.226 m . The 3D Flood Models produced within the scope of the study visualized that the susceptibility is not spatially homogeneous; on the contrary, it is concentrated in "topographic trap" areas such as the Kurbağalidere Basin and Yenisahra, which act as depressions. It has been demonstrated that the flow descending rapidly from high elevations (such as Acıbadem) becomes trapped in downstream points where the slope decreases and in valley floors, transforming the susceptibility from a spatial spread

into a volumetric threat, falling into the "Flash Flood" susceptibility group defined in the literature.

The calculated maximum water depth and flow velocity data constitute a significant pressure factor on existing infrastructure and transportation networks. Model outputs indicate that strategic transportation axes, especially E-5 highway connection points, metro entrances, and valley floors, are under risk of flooding if existing drainage lines prove insufficient.

Conversely, in regions containing green areas (Validebağ Grove, Yoğurtçu Park, and the coastal strip), CN values remained in the 60-70 range, providing concrete data that these areas act as "natural sponges" within the urban fabric, playing a vital role in flood attenuation and alleviating the drainage load. In light of these numerical findings and susceptibility maps, a holistic management strategy is required to increase the flood resilience of the Kadıköy district. By incorporating neighborhood-scale mapping, this study provides a concrete baseline for local decision-makers to prioritize infrastructure interventions. The results indicate that "Sponge City" strategies and the expansion of stormwater drainage capacities should be focused urgently on the Osmanağa depression and the downstream corridors of Kurbağalıdere, where water depths were calculated to reach critical thresholds. Ultimately, the integration of administrative boundaries with physical flood models transforms abstract hydrological data into actionable urban planning tools, identifying exactly which residents and critical transportation axes, such as those near the E-5 highway and metro entrances, are at the highest risk during extreme 100 year rainfall events.

Finally, it is recommended to transform the static model developed within the scope of this thesis through the integration of macro algorithms and GIS into a dynamic early warning system fed by IoT-based water level sensors in the future, and to implement smart city applications that direct traffic in risky underpasses.

The methodology and results presented in this study are supported by open source tools and custom developed algorithms. To ensure the reproducibility of the flood hazard analysis and the GeoAI validation process, the QGIS project files and the Random Forest classification scripts are made publicly available.

GitHub Repository:

<https://github.com/beyzabilgepayal/Flood-Susceptibility-Analysis-Kadikoy>

6. REFERENCES

- Breiman, L.** (2001). Random Forests. *Machine Learning*, 45(1), 5-32.
- Chow, V. T.** (1959). *Open-Channel Hydraulics*. New York: McGraw-Hill Book Company.
- Copernicus.** (2021). *Copernicus Global Digital Elevation Model (GLO-30)*. European Space Agency (ESA). Retrieved from <https://opentopography.org>
- Cronshey, R.** (1986). *Urban Hydrology for Small Watersheds (TR-55)*. Washington, DC: USDA Soil Conservation Service, Engineering Division.
- EEA (European Environment Agency).** (2018). *Corine Land Cover (CLC) 2018, Version 2020_20u1*. Copenhagen: European Environment Agency.
- Gazete Kadıköy.** (2014, August 14). *Kadıköy denizi tehlike saçıyor*. Retrieved January 13, 2026, from <https://www.gazetekadikoy.com.tr>
- Gazete Kadıköy.** (2019a, August 17). *İstanbul'da sel: Kadıköy'de metrekaresine 30 kg yağış!* Retrieved January 13, 2026, from <https://www.gazetekadikoy.com.tr>
- Gazete Kadıköy.** (2019b, August 21). *En fazla yağmur Kadıköy'e düştü*. Retrieved January 13, 2026, from <https://www.gazetekadikoy.com.tr>
- Gigovic, L., Pamucar, D., Bajic, Z., & Drobnjak, S.** (2017). Application of GIS-Interval Rough AHP Methodology for Flood Hazard Mapping in Urban Areas. *Water*, 9(6), 360.
- GRASS Development Team.** (2024). *Geographic Resources Analysis Support System (GRASS GIS) Software, Version 8.3*. Open Source Geospatial Foundation.
- Kadıköy Municipality.** (n.d.). *Kadıköy'de sel ve su baskını önlemleri çalışmaları sürüyor* [Photograph]. Retrieved January 13, 2026, from <https://www.kadikoy.bel.tr>
- Kourgialas, N. N., & Karatzas, G. P.** (2011). Flood management and a GIS modelling method to assess flood-hazard areas—a case study. *Hydrological Sciences Journal*, 56(2), 212-225.
- Manning, R.** (1891). On the flow of water in open channels and pipes. *Transactions of the Institution of Civil Engineers of Ireland*, 20, 161-207.

- MGM (Turkish State Meteorological Service).** (2024). *Historical Precipitation Data for Göztepe Station (Station No: 35919)*. Ankara: General Directorate of Meteorology.
- Mishra, S. K., & Singh, V. P.** (2003). *Soil Conservation Service Curve Number (SCS-CN) Methodology*. Springer Science & Business Media.
- QGIS Development Team.** (2024). *QGIS Geographic Information System*. Open Source Geospatial Foundation Project. <http://qgis.osgeo.org>
- Schumann, G. J.-P., & Bates, P. D.** (2018). The Need for a High-Accuracy, Open-Access Global DEM. *Frontiers in Earth Science*, 6, 225.
- Tehrany, M. S., Pradhan, B., & Jebur, M. N.** (2014). Flood susceptibility mapping using a novel ensemble weights-of-evidence and support vector machine models in GIS. *Journal of Hydrology*, 512, 332-343.
- Wang, Z., Lai, C., Chen, X., & Yang, B.** (2015). Flood hazard risk assessment model based on random forest. *Journal of Hydrology*, 527, 1130-1141.
- Yilmaz, A. G.** (2023). Flood Susceptibility Mapping Using GIS and Machine Learning: A Case Study of Istanbul. *Natural Hazards*, 115, 231-250.